Underactuated Robotics

Algorithms for Walking, Running, Swimming, Flying, and Manipulation

Russ Tedrake

Last modified 2024-5-30.
How to cite these notes, use annotations, and give feedback.

Note: These are working notes used for a course being taught at MIT. They will be updated throughout the Spring 2024 semester. Lecture videos are available on YouTube.

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Appendix E
Miscellaneous

E.1 How to cite these notes

Thank you for citing these notes in your work. Please use the following citation:

Russ Tedrake. *Underactuated Robotics: Algorithms for Walking, Running, Swimming, Flying, and Manipulation (Course Notes for MIT 6.832).* Downloaded on [date] from https://underactuated.csail.mit.edu/

```
@book{underactuated,
  title        = "Underactuated Robotics",
  subtitle     = "Algorithms for Walking, Running, Swimming, Flying, and Manipulation",
  howpublished = "Course Notes for MIT 6.832",
  author       = "Tedrake, Russ",
  year         = 2023,
  url          = "https://underactuated.csail.mit.edu",
}
```

E.2 Annotation tool etiquette

My primary goal for the annotation tool is to host a completely open dialogue on the intellectual content of the text. However, it has turned out to serve an additional purpose: it's a convenient way to point out my miscellaneous typos and grammatical blips. The only problem is that if you highlight a typo, and I fix it 10 minutes later, your highlight will persist forevermore. Ultimately this pollutes the annotation content.

There are two possible solutions:

- you can make a public editorial comment, but must promise to delete that comment once it has been addressed.

- You can join my "editorial" group and post your editorial comments using this group "scope".

Ideally, once I mark a comment as "done", I would appreciate it if you can delete that comment.

I highly value both the discussions and the corrections. Please keep them coming, and thank you!

E.3 Some great final projects

Each semester students put together a final project using the tools from this class. Many of them are fantastic! Here is a small sample.

Spring 2024 Outstanding Project Awards:

- Acceleration Constrained Quadrotor Trajectory Optimization in Dense Obstacle Fields Using Graphs Of Convex Sets by Michael Tibbs
- Collision Free Trajectory Optimization for Control of Fixed-Wing UAVs using Graph of Convex Set by Steve Nomeny
- Co-training for Visuomotor Policy Learning by Abhinav Agarwal and Adam Wei
- Graph of Convex Sets for Trajectory Optimization on Spot by Johannes Ihle, Aileen Liao, and Lukas Molnar
- Satellite Attitude Planning using Semidefinite Programming by Brandon Eickert and Shreeyam Kacker
- Here is a playlist containing all of the public projects from the term.

Spring 2023:

- Trajectory optimization for swing-up of multi-link cart-pole by Siro Corsi
- SalmonBot by Erick Fuentes
- Cooperative Shared-Load Carrying by Quadrotors by Seiji Shaw and Tommy Cohn
- Flying bird robot by Quang Kieu
- QP inverse dynamics for Spot with an arm by Namir Jawdat
- Staff Spinning Using Trajectory Optimization by Shruti Garg
- GCS as a policy by Savva Morozov

Spring 2022:

- Bowling Strikes with Trajectory Optimization by Benjamin Qi
- Quantum Control via Constrained Trajectory Optimization by Shantanu Jha and Shoumik Chowdhury
- Trajectory Optimization of Simple Skateboarding Tricks through Contact by Michael Burgess
- Contact-Aware Controller Design via Alternating Optimization by Richard Li and Timur Garipov

- Certifying Stability for Magnetically Actuated Satellites by Alex Meredith
- Quadrotor Trajectory Optimization with Stochastic Wind Disturbance by Mason Darveaux
- Contraction Metrics-based Stability Analysis via SOS Programming by Sunbochen Tang
- Brachiation Trajectory Optimization by Kaleb Blake and John Flynn

Spring 2021:

- Linearized MPC for a kinematic bicycle model by Mike Schoder. [code]
- Trajectory Optimization of Interplanetary Travel using Gravity Assist Maneuver
- Differential Flatness Trajectory Stabilization of Quadrotors by Charles Vorbach
- Trajectory Planning of a Quadrotor in Random Environments by Susan Ni, Christian Schillinger, and Max Thomsen

Spring 2020:

- Collision Free Mixed Integer Planning for Quadrotors Using Convex Safe Regions by Bernhard Paus Græsdal
- Pancake flipping via trajectory optimization by Charles Dawson
- Dynamic Soaring by Lukas Lao Beyer
- Acrobatic Humanoid by Matt Chignoli and AJ Miller
- Trajectory Optimization for the Furuta Pendulum by Samuel Cherna and Philip Murzynowski

Even before we started posting project presentations on YouTube, some great projects from class have turned into full publications. Here are a few examples:

- Feedback Control of the Pusher-Slider System: A Story of Hybrid and Underactuated Contact Dynamics
- Harnessing Structures for Value-Based Planning and Reinforcement Learning
- The famous Mini Cheetah backflip in Mini Cheetah: A Platform for Pushing the Limits of Dynamic Quadruped Control started as an underactuated project.

E.4 Please give me feedback!

I'm very interested in your feedback. The annotation tool is one mechanism, but you can also comment directly on the YouTube lectures, or even add issues to the github repo that hosts these course notes.

I've also created this simple survey to collect your general comments/feedback.

